XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	COM103			
Course Name:	Electronics Engineering			
Question Paper Setter:	Assoc Prof Ir Ts Dr Wong Shen Yuong			
Academic Session:	2022/04	Question Paper:	A ☑	B ☐
Total No. of Pages:	5	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	Calculator			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 5 Questions.

   Answer **ALL** questions.

2. Your answers must be handwritten.

3. Please follow the requirement of each question and paste on the downloadable answer book provided.

4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

NOTE: ALL ANSWERS IN THE FORM OF DECIMAL (0.000), NOT fraction ($\frac{X}{Y}$).

Question 1 [15 marks]

(CLO1, PLO1, C3)

(a) Explain an intrinsic semiconductor. [3 marks]

(b) Provide ONE (1) example for (a). And then, draw a diagram to show how it is formed. [8 marks]

(c) State what are the majority and minority carriers of n-type semiconductor. [4 marks]

Question 2 [19 marks]

(CLO1, PLO 1, C3)

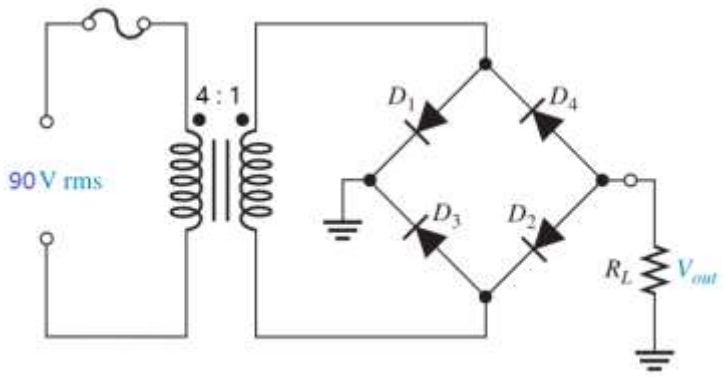


Figure 1

Consider the circuit in Fig. 1,

(a) What type of circuit is this? [3 marks]

(b) Calculate the peak secondary voltage. [4 marks]

(c) Calculate the peak output voltage. [4 marks]

(d) Sketch the output voltage waveform across R_L. [4 marks]

(e) What is the PIV for each diode? [4 marks]

Question 3 [10 marks]

(CLO1, PLO 1, C3)

Draw the output waveforms for all the 2 circuits in Fig.2 (a) to (b) showing all important values. Assume the RC time constant is much greater than the period of the input. [10 marks]

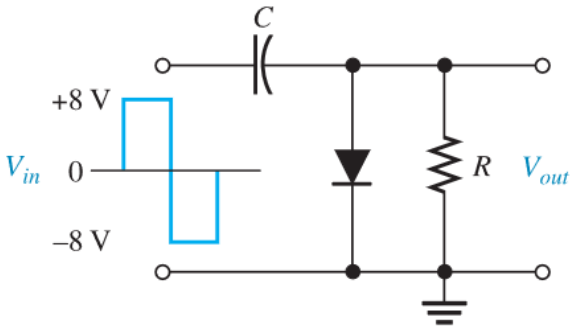


Figure 2(a)

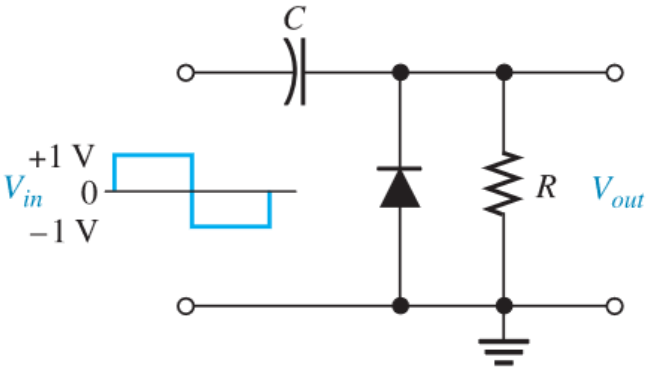


Figure 2(b)

Question 4 [18 marks]

(CLO 2, PLO2, C4)

(a) Under what bias condition is a varactor operated? [2 marks]

(b) What is the zener impedance if the zener diode voltage changes from 3.49 V to 4.99 V when the current changes from 5.00 mA to 10.0 mA? [4 marks]

(c) A 1N756 is a zener diode (8.2 V at 25° C) with a positive temperature coefficient of 5.8 mV/°C. What is the output voltage if the temperature rises to 65° C? [4 marks]

(d) Referring to Figure 3, assume a zener diode with $Z_Z = 30\Omega$ and $V_Z = 12\text{V}$ at 30 mA. What is the value of R to make $I_Z = 40mA$? [8 marks]

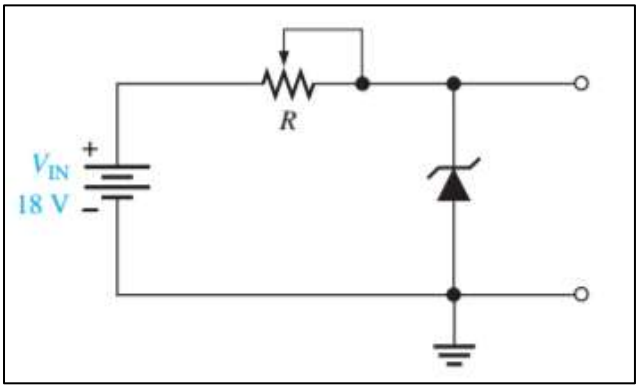


Figure 3

Question 5 [38 marks]

(CLO2, PLO 2, C4)

(a) Write down the two types of BJTs, and draw the symbols for the two types. [6 marks]

(b) The BJT is a three terminal device. Show and name the three terminals. [3 marks]

(c) Referring to Fig. 4, the circuit has the parameters: V_{BB} = 8V, R_B = 440 kΩ, R_C = 4 kΩ, Vcc = 20V, β = 200. Solve for I_C, I_E, I_B, V_{CB}, V_{CE}, and V_{BE}. [18 marks]

(d) Referring to Fig. 4, determine whether or not the transistor is saturated. Given $V_{CE(sat)}$ = 0.2V. [11 marks]

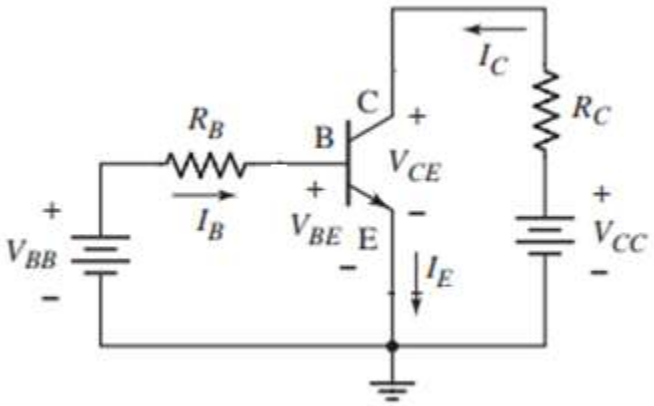


Figure 4

- END OF QUESTIONS -